

China Strategies in the Automotive Industry

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1. Introduction

For three decades, the joint venture represented the dominant model for building automobiles at scale in China: foreign technology was exchanged for market access, with a state-owned enterprise serving as the indispensable domestic partner (Holmes et al., 2015). That model is now unraveling. Electric vehicles accounted for approximately 55% of new vehicle sales in China in 2025, while Chinese manufacturers supplied around 60% of global EV sales that year (IEA, 2026). Firms that once relied heavily on foreign partners are losing domestic market share at an accelerating pace (SAIC Motor, 2025), whereas independent Chinese manufacturers have emerged as global leaders (Whitfield & Wuttke, 2026). The same conditions are eroding one organizational model while validating others.

These diverging trajectories are not coincidental. State-owned enterprises built around mandatory joint ventures have struggled to maintain their historical advantages, while privately controlled firms have surpassed established Western manufacturers in several segments. Private ownership may facilitate faster strategic adaptation, whereas state-owned firms must balance commercial objectives with broader political considerations (Shleifer & Vishny, 1994; Megginson & Netter, 2001). The distinction extends beyond firm-level performance: the European Commission's 2024 countervailing duties impose 35.3% on SAIC against 17.0% on BYD and 18.8% on Geely, transforming governance structures into a variable of international trade policy (European Commission, 2024).

This paper investigates which governance and financing structure is most likely to create long-term shareholder value in a technology-intensive, high-growth market. It compares three models through which Chinese automakers have achieved global scale: the joint-venture-based, the vertically integrated, and the acquisition-driven model. The analysis explores how each shapes access to technology, capital, and scale, and how risk is allocated among the firm, its foreign partners, and the state.

SAIC Motor represents a state-controlled, joint-venture-centric model majority-owned by Shanghai SASAC. This structure generated stable returns for over two decades but has become increasingly exposed as foreign-brand vehicles have lost competitiveness, coinciding with an 88% decline in net profit in 2024 (SAIC Motor, 2025). BYD exemplifies organic growth through deep vertical integration, developing proprietary battery capabilities ahead of state policy to lead the global NEV market (BYD, 2025; Whitfield & Wuttke, 2026). Geely pursued growth through acquisitions, most notably Volvo Cars in 2010, accessing advanced platforms and brand equity faster than organic development would have permitted (Balcet et al., 2012; Luo & Tung, 2007).

The analysis proceeds along four dimensions: governance, financing, capability building, and long-term viability, drawing on audited annual reports, primary regulatory sources, and peer-reviewed literature. The central argument is that governance structure, more than initial technological endowment, shapes a firm's capacity to adapt under rapid technological change and determines whether capital is deployed strategically for future growth or defensively in response to structural decline.

2. Governance

2.1 SAIC Motor: State Ownership and Political Governance

SAIC Motor Corporation Limited operates under a two-tier state ownership pyramid, a structure frequently used in civil-law jurisdictions to maintain government control over listed firms (Bortolotti & Faccio, 2009). Shanghai SASAC holds 63.71% of SAIC's shares through its wholly owned subsidiary Shanghai Automotive Industry Group, and including other state blockholders such as China COSCO Shipping (5.91%) and Yuejin Motor Group (3.60%), the cumulative state share reaches approximately 73% (SAIC Motor, 2026). Although listed on the Shanghai Stock Exchange (code: 600104), ultimate authority over board appointments and strategic capital allocation remain with SASAC.

The formal role of the Chinese Communist Party is explicitly integrated into this structure. SAIC's 2025 Annual Report documents adherence to the principle of "two consistencies", which balances Party leadership with the development of a modern enterprise system (SAIC Motor, 2026, p. 35), making Party governance a constitutive element of formal authority rather than merely an informal influence.

Shleifer and Vishny (1994) argue that control rights, rather than cash-flow rights, determine resource allocation, enabling political actors to pursue objectives beyond shareholder value maximization. Applied to SAIC, this suggests that SASAC ownership and Party oversight systematically steer corporate decisions toward political and strategic state objectives. Empirical support comes from Fan et al. (2007), who find that politically connected Chinese firms underperform unconnected peers by 18% in three-year post-IPO returns and exhibit slower sales and earnings growth.

In 2025, SAIC restructured its governance to comply with China's revised Company Law, abolishing the Board of Supervisors and consolidating its functions within a strengthened Audit Committee. These adjustments signal formal modernization but leave the core own-

ership structure and Party leadership mandate unchanged. As long as ultimate control remains in state hands, procedural reform alone is insufficient to realign incentive structures, a conclusion consistent with Megginson and Netter (2001), who find that mixed-ownership enterprises typically do not outperform fully privatized firms.

2.2 BYD: Founder-led Private Governance

BYD is controlled by its founder Wang Chuanfu, who has served simultaneously as Chairman and CEO since the company's founding in 1995 and directly holds approximately 17% of shares (BYD, 2025). Together with co-founders Lu Xiangyang, who holds around 14% and through Youngy Investment and serves as Non-Executive Director on the board and Xia Zuo-quan who holds roughly 3%, the founding team commands more than one third of voting rights under a one-share-one-vote structure (BYD, 2025). Other significant shareholders, including Berkshire Hathaway and BlackRock, with approximately 3.0% each have no operational decision-making authority and are not represented on the board. The company is privately organised; state subsidies amounted to RMB 10.4 billion in 2024 but confer no control rights (BYD, 2025). A party committee with strategic directive authority as structurally embedded in state-affiliated enterprises does not exist at the operational level. Unlike SAIC Motor, which historically built its market position on state-mandated joint ventures with Western OEMs, BYD emerged as an independent private manufacturer without JV dependency a structural freedom that shaped its governance model from the outset (Yuan & Brasó Broggi 2025).

This structure has direct consequences for decision-making speed. Concentrated founder control as embodied by Wang structurally enables rapid, unilateral strategic decisions without the consensus requirements of joint-venture boards or state-owned enterprise oversight (Zhang et al. 2025). Strategic decisions including the complete exit from combustion engine production in 2022, the establishment of BYD Semiconductor as an independent subsidiary, and the formal restructuring of component divisions into the FinDreams Group were implemented unilaterally, without external shareholder approval or political authorization (BYD, 2025; Wang, Zhao & Ruet 2022). Zhang et al. (2025) provide empirical support for this mechanism: founder control in Chinese high-tech firms promotes breakthrough innovation because it shields against short-term capital market pressure and concentrates technical expertise with decision-making authority. Wang Chuanfu's background as a trained chemist corresponds to the ideal type of founder-technologist identified in the paper (Zhang et al. 2025).

A key advantage arises from this ownership structure: because Wang Chuanfu retains control and no broad shareholder base demands short-term profit distributions, BYD can consistently reinvest in long-term projects a discipline that would be difficult to sustain under dispersed ownership with quarterly earnings pressure. BYD's long-term capital allocation reflects a consistent prioritization of technological capability over short-term profit distribution. In 2024, BYD invested approximately RMB 54.2 billion in research and development, a year-on-year increase of 35.7% exceeding net profit of RMB 40.3 billion by a factor of 1.3 (BYD, 2025). This pattern persisted into 2025, when R&D expenditure of RMB 63.4 billion again exceeded net profit of RMB 32.6 billion (BYD, 2026). Because founder ownership concentrates control and aligns managerial incentives directly with shareholder interests, it eliminates the principal-agent conflict inherent in externally managed firms a governance quality that capital markets recognize and reward with higher firm value (Jensen & Meckling 1976). At the institutional level, Allen et al. (2024) provide empirical confirmation for the Chinese context: state-controlled listed firms consistently underperform their privately governed counterparts, with weak corporate governance identified as the primary driver. BYD's founder-controlled, state-free ownership structure thus represents the governance model that both theory and evidence identify as superior (Allen et al. 2024).

2.3 Geely: M&A-Driven Private Governance

Geely's governance model represents a third distinct archetype in Chinese automotive development: a privately held conglomerate that deploys cross-border M&A as its primary capability-building mechanism. Zhejiang Geely Holding Group (ZGH) is wholly owned by founder and chairman Li Shufu; his aggregate attributable interest in the publicly listed Geely Automobile Holdings (HK: 0175) is held through Geely Holding (39.20%) and Proper Glory Holding Inc. (25.24%), together approximately 41% of issued share capital as at 31 December 2024, with Li Shufu retaining sole private ownership of the unlisted ZGH apex entity (AR2025, p.76). This concentrated founder-ownership structure, unencumbered by state committee approval processes or joint-venture partner consent, confers the strategic agility to execute rapid, large-scale foreign acquisitions.

Luo and Tung's (2007) springboard perspective provides the appropriate theoretical anchor: emerging-market firms use cross-border acquisitions of developed-market assets as springboards to overcome latecomer disadvantages in technology, brands, and management competence. ZGH has operationalised this logic across a sequenced portfolio of transactions: the USD 1.5 billion acquisition of Volvo Cars from Ford Motor Company in 2010, the acquisition of LEVC (London Electric Vehicle Company) in 2013, a 51% controlling stake in Lotus Cars in May 2017, and the formation of a 50:50 joint venture with Mercedes-Benz for the

Smart brand in January 2020, with registered capital of approximately USD 0.77 billion (RMB 5.4 billion).

The governance challenge of administering this multi-brand portfolio is resolved through differentiated integration rather than centralised control. Zheng, Noorderhaven and Du (2022) document how Geely managed the Volvo relationship through four sequential stages, distancing, balancing, building, and diversifying, driven by legitimacy pressures from both host-country stakeholders and acquired-firm employees. This arm's-length philosophy preserved Volvo's engineering culture while enabling capability transfer, most concretely through the jointly developed Compact Module Architecture (CMA) platform; analogous logic grants Lotus its British performance identity and the Smart JV formally co-equal board representation.

By contrast, SAIC Motor's state-directed governance constrains strategic autonomy through its SOE ownership structure, while BYD's founder-led model pursues capability through a vertically integrated model including battery and semiconductor production. Geely's approach trades the cost efficiency of vertical ownership for the speed of strategic asset acquisition and the international legitimacy conferred by established Western brands.

2.4 Comparative Synthesis: Who Controls, and What They Optimize

The three models differ fundamentally in who controls the firm and what that controller optimizes. SAIC's governance subjects capital allocation to a dual principal: Shanghai SASAC pursues state-directed objectives alongside shareholder value, while the formally codified Party leadership mandate extends political control beyond ownership into strategic decisions (Shleifer & Vishny, 1994). Megginson and Netter (2001) establish that mixed-ownership structures systematically underperform privately governed peers; Fan et al. (2007) quantify the cost at approximately 18% in three-year post-IPO returns for politically connected Chinese firms. Neither BYD nor Geely faces this constraint: both operate under private founder ownership with a commercial objective function, enabling the unilateral strategic decisions documented in Sections 2.2 and 2.3.

Within the private models, governance complexity distinguishes the two. BYD's concentrated founder control eliminates the principal-agent conflict between ownership and management (Jensen & Meckling, 1976), producing the fastest decision cycle of the three firms. Geely achieves comparable acquisition agility at the apex level but introduces coordination costs across a multi-brand portfolio operating under differentiated governance arrangements (Zheng et al., 2022). The EU countervailing duties translate this hierarchy into a measurable trade variable 35.3% on state-governed SAIC against 17.0% on BYD and 18.8% on Geely confirming that in globally contested industries, governance structure is no longer merely an

internal organizational choice but a first-order determinant of competitive position (European Commission, 2024).

3. Financing

3.1 SAIC Motor: The JV Capital Model and Its Structural Limits

SAIC Motor's historical financing model was capital-light by design. Under China's quid pro quo policy, foreign automakers were required to form 50/50 joint ventures to gain market access, obligating partners such as Volkswagen and General Motors to share costs and risks (Holmes et al., 2015). Because the foreign partners contributed core technology, vehicle platforms, and brands, SAIC earned stable equity income for decades without funding proprietary vehicle development of its own. The arrangement also delivered scale: even after a 20% decline in 2024, SAIC sold roughly 4.0 million vehicles, keeping it among China's largest automakers by volume (SAIC Motor, 2025). That scale, however, had long rested on JV brands, whose contribution is now receding as own-brand vehicles approach 60% of sales.

However, these rents came at a financing cost. The steady JV dividends plausibly reduced the pressure to reinvest retained earnings into proprietary R&D, channelling capital toward distribution rather than independent vehicle development (Howell, 2018). This capital-allocation pattern left SAIC structurally exposed once the JV income itself began to erode.

The structural vulnerability became apparent in 2024. SAIC-GM's annual sales collapsed by 56.54% to 435,007 units, triggering large impairment charges that sharply reduced SAIC's investment income from the joint venture. Total operating revenue declined by 15.73% to RMB 627.59 billion, reported net profit plummeted 88.19% to RMB 1.67 billion, and net profit excluding non-recurring items was negative at RMB 5.41 billion. Weighted average ROE fell to 0.58%, its lowest level in recent years, down from 4.98% in 2023 and 5.84% in 2022 (SAIC Motor, 2025). SAIC remained contractually bound to absorb these losses while lacking the operational control to restructure the foreign-led entities. SAIC reported a headline recovery in 2025, with net profit rising to RMB 10.11 billion, but this largely reflects the non-recurrence of the prior year's impairments. Net profit excluding non-recurring items recovered to RMB 7.42 billion, still below the 2023 level of RMB 10.04 billion, and weighted average ROE reached only 3.43% (4.98% in 2023), indicating financial stabilization rather than a structural recovery of the JV-centric business model (SAIC Motor, 2026).

A second dimension of SAIC's financing dependence is its reliance on state-mediated capital, formally quantified by the European Commission's 2024 anti-subsidy investigation. The Commission imposed a definitive countervailing duty of 35.3% on SAIC's battery electric vehicle exports, more than double the rates applied to BYD (17.0%) and Geely (18.8%), after documenting extensive preferential financing through state-backed loans, bonds, and asset-backed securities, partly on the basis of "facts available," since the group withheld key financing information (European Commission, 2024). The differential serves as a regulatory indicator of SAIC's disproportionate reliance on state support relative to its private-sector peers.

3.2 BYD: Vertical Integration Financing

BYD's financing model differs fundamentally from debt-intensive growth strategies: the company funds its expansion primarily from ongoing operations. In fiscal year 2024, BYD generated operating cash flow of RMB 133.5 billion on revenues of RMB 777.1 billion (+29% year-on-year) and net profit of RMB 40.25 billion (+34%) (BYD, 2025). Cash reserves stood at RMB 154.9 billion at year-end, while interest-bearing liabilities declined to RMB 28.58 billion representing only 4.9% of total liabilities (BYD, 2025). Internal financing is supplemented by state subsidies, which amounted to RMB 10.41 billion in 2024, equivalent to approximately 26% of annual net profit (BYD, 2025).

The central advantage of this model lies in independence from capital markets. Since BYD does not need to finance its investments through external debt or equity issuances, there is no pressure to prioritise short-term profitable projects over long-term research initiatives. In 2024 alone, R&D expenditures of RMB 54.2 billion exceeded annual net profit of RMB 40.25 billion, a structural reinvestment pattern that would be difficult to sustain under dispersed ownership with quarterly earnings pressure (BYD, 2025; Zhang et al. 2025). A further structural strength is margin resilience: gross margin rose to a record 19.4% in 2024 despite intense price competition in the Chinese automotive. BYD's financial statements confirm this directly: vertical integration has steadily expanded research capacity while reducing supplier dependence, generating an internal cash flow base sufficient to fund investment without recourse to external debt (BYD, 2025). The resulting net cash position eliminates significant interest obligations, conferring a liquidity buffer that externally financed competitors structurally cannot replicate.

However, the model has structural limits. First, growth speed is constrained by internally generated liquidity. In the early growth phase, BYD depended on external capital: in 2008, Berkshire Hathaway acquired approximately a 10% stake, providing BYD with access to

international capital and reputation (BYD, 2025; Whitfield & Wuttke, 2026). Second, subsidy dependence carries political risk: state subsidies currently amount to approximately one quarter of net profit and are tied to the continuation of Chinese industrial policy (BYD, 2025). A reduction or elimination of these funds would immediately constrain investment capacity. This risk should be qualified, however: Jiang & Xu (2023) show, based on 196 Chinese NEV companies, that state subsidies primarily increase the quantity of innovations rather than their quality. BYD's structural innovation depth, sustained by vertical integration and proprietary R&D capabilities, would remain largely intact in the event of a subsidy reduction. Third, and most fundamentally, the reliability of a cash-flow-based model depends on stable operational profitability, making it a limited tool for valuing companies in dynamic, high-growth phases.

3.3 Geely: Acquisition Financing

The 2010 Volvo acquisition is the paradigm case of leveraged, state-assisted cross-border financing in Chinese automotive history. Balcer, Wang and Richet (2012) document the structure: ZGH paid USD 1.5 billion for 100% of Volvo Cars, including a USD 200 million seller note from Ford, with roughly USD 1.1 billion provided by a coalition of Chinese state-linked entities and a further USD 200 million from China Construction Bank (London). Including operational reserves, total capital raised approached USD 2.7 billion, with bank credit from the Bank of China and Export-Import Bank of China alongside CCB. The consequences for the balance sheet were severe: group debt surged from ~USD 2.35 billion (2009) to USD 10.5 billion (2010), pushing the debt-to-capital ratio to 73.5%, well above the industry high of ~60%, while ZGH absorbed Volvo's pre-existing USD 3.5 billion in obligations.

The strategic logic was speed. Organic development of equivalent brand, platform, and market assets would have required decades; the acquisition compressed that timeline to a single transaction. The thesis was vindicated by Volvo Cars' IPO on Nasdaq Stockholm in October 2021, which valued the company at approximately USD 18.5 billion, more than ten times the 2010 acquisition price, while ZGH retained a ~75–80% ownership stake, realising substantial unrealised gains.

Geely's subsequent portfolio, however, carries material refinancing risk. Polestar (Nasdaq: PSNY) recorded a net loss of USD 1.17 billion in FY2023, received a going-concern qualification from auditor Deloitte, and held only USD 103.1 million in cash at end-2024; analysts estimate a further USD 2 billion in capital requirements through 2028, with ZGH committed as group-level guarantor. Lotus Technology has accumulated a deficit of approximately USD 3.2 billion as of FY2025, with revenue falling 44% year-on-year in FY2025.

The contrast to peer financing models is instructive. BYD's vertically integrated model builds capability organically, avoiding acquisition debt but constraining the speed and geographic breadth of expansion. SAIC Motor's joint venture model shares capital requirements with foreign partners, limiting balance-sheet exposure while also constraining strategic independence. Geely's leverage-for-speed approach carries the highest financing risk of the three models, with group-level debt-servicing capacity remaining structurally dependent on Volvo's sustained profitability.

3.4 Comparative Synthesis: Risk Allocation Across the Three Models

The three financing models differ structurally in their primary capital source, risk allocation, and investment incentives. SAIC's historical JV model externalised capital risk by splitting equity 50/50 with Western partners, enabling rapid capacity scaling without proportionate balance-sheet exposure but the same design made R&D autonomy structurally impossible and created dependency on partner profitability that collapsed once SAIC-GM's sales fell 56.5% in FY2024. BYD's self-financing model keeps investment decisions entirely internal: R&D expenditure of RMB 54.2 billion in FY2024 exceeded net profit of RMB 40.3 billion, a discipline that dispersed ownership with quarterly earnings pressure would structurally resist (BYD 2025). Geely's acquisition model trades financial stability for capability speed leveraging brand equity and platform access from Volvo and Lynk&Co at the cost of rising gearing, which reached 19.8% by end-FY2025 (Geely Annual Report 2025).

The comparative data in Table 1 reveal two cross-sectional divergences. On return on equity, BYD's 21.7% in FY2024 versus SAIC's 0.58% and Geely's 19.2% reflects the efficiency advantage of vertical integration when demand is favourable; its compression to 13.2% in FY2025 under margin pressure indicates sensitivity to the price cycle. On gearing trajectory, both BYD and Geely shifted toward net debt positions in FY2025 (BYD from -36% to +25%; Geely from 8.8% to 19.8%), driven by global factory expansion and the Lynk&Co acquisition respectively. Allen et al. (2024) confirm that private governance generates structurally higher long-run returns than state-controlled firms. The ROE divergence between BYD (21.7%) and SAIC (0.58%) in FY2024 quantifies this directly. The FY2025 gearing shift at both BYD and Geely signals, however, that self-financing resilience is not unconditional: it holds only as long as the operating margins that fund it.

Table 1: Comparative Financial and Structural Profile

Metric	SAIC Motor (JV / SOE)		BYD (Private / Vertical)		Geely (Private / M&A)	
	FY2024	FY2025	FY2024	FY2025	FY2024	FY2025
Revenue (RMB bn)	627.6	656.2	777.1	804.0	240.2	345.2
Gross Margin	9.4%	n.a.	19.4%	17.7%	15.9%	16.6%

Net Profit (RMB bn)	1.7	10.1	40.3	32.6	16.8	16.9
R&D Expense (RMB bn)*	35.2	33.6	54.2	63.4	10.4	17.6
R&D Intensity (% Rev.)	5.6%	5.1%	7.0%	7.9%	4.3%	5.1%
Operating Cash Flow (RMB bn)	69.3	n.a.	133.5	59.1	26.5	47.3
Return on Equity (ROE)	0.58%	3.43%	21.7%	13.2%	19.2%	18.2%
Net Gearing	n.a.		-36%	+25%	8.8%	19.8%
EU Anti-Subsidy Tariff	35.3%		17.0%		18.8%	
Primary Capital Source	<i>State equity + JV dividends (hist.)</i>		<i>Operating cash flow (retained earnings)</i>		<i>Debt (acquisition-financed)</i>	
Risk Bearer	<i>State / JV partners (50/50)</i>		<i>Equity holders (founder-led)</i>		<i>Debt providers</i>	

* SAIC R&D total includes JV contributions. SAIC standalone FY2024: RMB 17.7 bn (2.8% of revenue). n.a. = not available in published report. Sources: SAIC AR 2024/2025, BYD AR 2024/2025, Geely AR 2024/2025.

4. Capability Building

4.1 SAIC Motor: Technology Transfer and the Capability Paradox

SAIC's capability development unfolded within the quid pro quo regime described in Section 3.1. Holmes et al. (2015) conceptualize this as an implicit tax on technology capital: since transferred technology becomes Chinese property, foreign firms had reason to withhold their most advanced capabilities. As the state-designated partner of Volkswagen and General Motors, SAIC stood at the center of this regime and among its primary beneficiaries (Bai et al., 2025; Holmes et al., 2015).

Bai et al. (2025) offer the most direct causal evidence on the policy's effects. Drawing on J.D. Power's Initial Quality Study for nearly all car models built in China, they estimate that quid pro quo affiliation contributed 8.3% of the total quality improvement in affiliated domestic models, with the gap narrowing from 65% higher malfunction rates in 2001 to 33% by 2014. Transmission occurred mainly through worker mobility and supplier networks, not formal patent transfer. Both SAIC Volkswagen and SAIC General Motors feature in the study, and SAIC's 2006 Roewe launch reflects the capability accumulation it enabled.

Howell (2018) documents a countervailing effect. Exploiting China's 2009 Phase 2 fuel economy standards as a quasi-experiment, she finds that JV-linked firms responded by reducing rather than upgrading quality: torque fell by 11% and prices by 15% relative to non-JV firms. The mechanism is a cannibalization dynamic: JV rents reduce the incentive to innovate autonomously, since upgrading own-brand technology competes with the profitable

foreign-brand income stream. Former Minister of Machinery He Guangyan called JV rents "like opium," addictive in the short term and detrimental to independent capability development (as cited in Howell, 2018, p. 1448).

Together, these studies describe a capability-building paradox: the JV model transferred production quality but suppressed the autonomous innovation required for next-generation capabilities. By 2014 SAIC had built far better vehicles under the Roewe and MG marques (Bai et al., 2025), yet lacked the proprietary battery, power electronics, and software platforms that would later define EV-era advantage. Inkpen and Beamish (1997) argue that joint ventures turn unstable once asymmetric learning erodes the knowledge-dependency sustaining them, a process underway by 2014 and fully visible by 2025, when SAIC-GM ran at 37% of capacity and Volkswagen's MEB plant at 14% (SAIC Motor, 2026). The fiscal-year 2025 annual report documents SAIC's response: NEV sales of 1.643 million units (+33.14%), what it calls the industry's first mass deployment of semi-solid batteries, and the SAIC SHANGJIE brand launched with Huawei (SAIC Motor, 2026). These steps are substantive but came well after BYD's comparable investments had secured a sustained market position.

4.2 BYD: Internal Build

BYD's capability building follows an internal logic that differs fundamentally from JV-based technology transfer or capability acquisition. Rather than entering battery technology as an automaker, BYD followed the reverse path: founded in February 1995 as Shenzhen BYD Battery Company Limited, the company first built manufacturing knowledge and experience as a supplier of lithium-ion batteries for consumer electronics before entering vehicle production in 2003 through the acquisition of the bankrupt state-owned Qinchuan Auto. The automotive entry thus drew on pre-existing cell chemistry expertise accumulated entirely outside the automotive sector a foundation that BYD subsequently applied to EV battery development from the mid-2000s onwards BYD began investing in automotive batteries not in response to state industrial policy, but ahead of it, thereby actively shaping the policy agenda (Whitfield & Wuttke 2026). Wang, Zhao & Ruet (2022) describe BYD's approach as "Specialised Vertical Integration": a selective integration concentrated on strategically complementary value chain stages, which deepens core competencies rather than diffusing them.

This model generates advantages that are structurally difficult to replicate. Teece, Pisano & Shuen (1997) show that sustainable competitive advantage arises from path dependently accumulated tacit knowledge embedded in organizational routines that cannot be transferred

through licensing or talent recruitment. BYD's decades of internal development produce precisely this kind of non-imitability. Moreover, each mastered competency opens access to the next. Whitfield & Wuttke (2026) documented how BYD leveraged existing capabilities from the consumer battery sector to build automotive specific knowledge incrementally from cell chemistry through battery packs and power electronics to vehicle software. Through this early positioning, BYD did not merely remain a technology follower but became a market architect whose development trajectory state subsidy programmes subsequently adopted (Whitfield & Wuttke 2026).

The model nevertheless has structural limits. First, internal capability building is time intensive. While a competitor can acquire a new capability through acquisition within months, organic learning requires years or decades a disadvantage during technological disruptions with compressed time windows. Second, the model is capital-intensive. Investments occur long before their commercial exploitation, increasing dependence on stable cash flow sources. Third, path dependence carries the risk of path rigidity: a paradigm shift in technology would partially devalue BYD's accumulated knowledge capital, as any redirection would require fundamental reorganisation (Teece, Pisano & Shuen 1997). Finally, simultaneous mastery of highly heterogeneous competency domains places significant demands on management: integrating diverse technological domains within a single firm generates organisational complexity that becomes harder to coordinate as the company grows.

4.3 Geely: Acquisition-based Capability Building

The 2010 Volvo acquisition gave ZGH instantaneous access to a cluster of strategic assets that would have taken Geely comparable decades to develop organically: Volvo's safety engineering expertise, three cutting-edge vehicle platforms, accumulated brand equity, and a global distribution network of over 2,400 dealer outlets. In the academic literature this is characterised as strategic asset-seeking FDI, Geely did not acquire Volvo for Swedish market access but for the engineering and institutional capabilities embedded in the brand, capabilities that then served as the basis for upgrading Geely's own products and processes.

The mechanism of absorption is documented in the four-stage integration model proposed by Zheng, Noorderhaven and Du (2022): distancing, balancing, building, and diversifying. The flagship output of the building stage is the jointly developed Compact Module Architecture (CMA) platform, deployed across Volvo's 40-series models, the Lynk & Co brand, and Geely-branded vehicles, concrete evidence that the transfer evolved from one-way absorption into bilateral co-development. That the deal catalysed rather than substituted for internal investment is confirmed by a 600% increase in Geely's R&D expenditure within seven years of acquisition, a trajectory that continued into the EV era: in H1 2025, Geely

Automobile Holdings reported R&D expenditure of approximately USD 1.02 billion (+21% year-on-year) (RMB 7.33 billion), equivalent to approximately 4.9% of H1 2025 revenue, confirming that external acquisition capability and internal innovation spend are complements rather than substitutes in Geely's model.

Konda, Slepnikov and Jin (2022) supply the quantitative foundation: longitudinal patent portfolio analysis reveals measurable convergence in the innovation outputs of Geely and Volvo over the post-acquisition period, distinguishing Geely-Volvo as a rare case of genuine co-creation among emerging-market acquirers rather than a one-directional extraction of developed-market knowledge.

The contrast with BYD's capability-building model is structurally significant. BYD's vertically integrated model, including battery and semiconductor production, follows a slower organic trajectory, but confers full proprietary control with no integration risk. Geely's acquisition route compressed the capability timeline decisively, yet its success remained contingent on sustaining the arm's-length governance conditions that protected Volvo's engineering culture; any deterioration of that relationship would have undermined the very assets the acquisition was designed to absorb.

4.4 Comparative Synthesis: Speed versus Depth in Capability Building

The three pathways reflect fundamentally different trade-offs between speed and depth of capability acquisition. SAIC's JV model delivered production quality fastest Bai et al. (2025) attribute an 8.3% quality gain to quid pro quo affiliation but created the capability paradox: JV rents suppressed autonomous innovation, producing a firm that mastered conventional assembly without ever building the battery, power electronics, or software platforms EV competition now demands. Geely compressed the timeline through acquisition and, uniquely among the three, converted external purchase into genuine co-development: four integration stages yielded the CMA platform, with measurable patent portfolio convergence confirming bilateral knowledge creation rather than one-way extraction (Zheng, Noorderhaven & Du 2022; Konda, Slepnikov & Jin 2022). BYD's organic path was the slowest but the deepest. Three decades of internal development from cell chemistry through power electronics to vehicle software produced the path-dependent tacit knowledge Teece, Pisano & Shuen (1997) identify as structurally non-imitable.

These structural origins are now legible in the competitive record. SAIC-GM at 37% capacity utilisation and VW's dedicated MEB plant at 14% in 2025 mark the endpoint of capability dependency hardening into capability deficit precisely the instability Inkpen and Beamish (1997) predict when asymmetric learning dissolves a JV's knowledge anchor. Geely's platform-based electrification demonstrates that absorbed capabilities can generate competitive

EV products, yet co-development dependencies on Volvo Cars and the Smart JV constrain unilateral decisions in ways BYD does not face. BYD's full proprietary command of battery chemistry, semiconductors, and vehicle software confers maximum platform autonomy and an iteration pace no partner-dependent architecture can replicate. Capability building is therefore not merely a record of how each firm learned it is the structural foundation that determines which model enters the EV era from a position of strength.

5. Long-term Viability

5.1 SAIC Motor: Joint Venture Erosion and the EV Transition

SAIC's long-term viability turns on a single transition: whether its own-brand electric vehicles can replace joint venture dividends as the group's profit engine before that engine fails. Sections 3.1 and 4.1 showed the JV model eroding structurally, its profitability collapsing in 2024 and its dedicated plants running far below capacity. The question is therefore not whether the JV model declines, but whether the own-brand business can scale fast enough to offset it.

One indicator suggests the pivot has commercial substance: the own-brand NEV growth noted in Section 4.1 was achieved without direct vehicle subsidies, recorded at zero for both passenger and commercial NEVs in the 2025 annual report (SAIC Motor, 2026). Unlike the JV rents that once flattered SAIC's accounts, this revenue stands on a commercial footing. The "Joint Venture 2.0" strategy reinforces the shift, repurposing idle JV capacity for own-brand EV production and turning a stranded liability into potential output (SAIC Motor, 2026).

Three constraints temper this. The price war compressing margins across the market falls hardest on own-brand volume producers, so rising units do not yet mean rising profit; SAIC's 2025 recovery rested largely on the non-recurrence of prior impairments rather than operational gains (Section 3.1). The export route that might widen margins is foreclosed by the European Commission's 35.3% countervailing duty (Section 3.1). And the pace of capital reallocation is bounded by the state ownership and Party leadership mandate of Section 2.1, which favour continuity over decisive redeployment.

SAIC's viability thus reduces to a race between declining JV rents and a still-thin own-brand margin. That the own-brand business is growing without direct subsidies makes the transition genuinely possible; yet the same state governance that sustained the JV model for three decades now constrains the speed at which SAIC can leave it. Success will depend less on technology, which SAIC increasingly possesses, than on whether its governance lets capital move as fast as the market.

5.2 BYD: Structural Leadership Under Margin Pressure

BYD has structurally completed the transition to electric mobility. Following the complete exit from combustion engine production in 2022, no ICE segment burdens capital allocation any further. For fiscal year 2025, the Annual Report shows more than one million exported vehicles for the first time, an increase of 140% year-on-year with a presence in 119 countries, structurally reducing BYD's dependence on the Chinese domestic market. For the fourth consecutive year, BYD was the global market leader in the NEV segment (BYD, 2026). Allen et al. (2024) show that externally listed Chinese private companies such as BYD achieve long-term capital market performance at global market level, while state owned companies structurally underperform. The capital increase of USD 5.6 billion in 2025, involving institutional long-term investors and sovereign wealth funds, signals sustained investor demand (BYD, 2026).

Structural risks nonetheless persist. The ongoing price war in the Chinese market leaves visible marks: gross margin fell from 19.4% (2024) to 17.7% (2025), and net profit declined by approximately 19% to RMB 32.6 billion indicating that vertical integration can dampen but not fully absorb systemic price pressure. In parallel, the leverage ratio shifted from a net cash surplus to a net debt position because of capital intensive global factory expansion, partially constraining the self-financing advantage described in Chapter 3 (BYD, 2026). For the export strategy, the EU countervailing duty of 17% with a combined tariff of 27% including the standard import duty remains a structural obstacle in the most profitable target market (EU Implementing Regulation 2024/2754).

5.3 Geely: Long-term Viability

Geely's FY2024 core performance was strong: Geely Automobile Holdings reported revenue of approximately USD 33.2 billion (+34% year-on-year) (RMB 240.2 billion) and net profit of approximately USD 2.32 billion (+240%) (RMB 16.8 billion), while Volvo Cars, the anchor acquisition, delivered SEK 400.2 billion in revenue at a 6.8% core EBIT margin, its second consecutive record year. The momentum has carried into 2025. In H1 2025, Geely Automobile Holdings reported revenue of approximately USD 20.9 billion (+27% year-on-year) (converted from RMB 150.3 billion at 7.20 CNY/USD), with total vehicle sales of 1,409,180 units (+47%), including a milestone: NEV sales reached 725,151 units (+126%), representing 51.5% of total volume, the first half-year in which electrified vehicles constituted a majority of Geely's sales mix. Adjusted operating profit for H1 2025 was approximately USD 0.92 billion (+102% year-on-year) (RMB 6.66 billion), with the reported net profit of approximately USD 1.29 billion (converted from RMB 9.29 billion) appearing down 14% only because H1 2024 contained a USD 1.07 billion one-off gain from the Horse

Powertrain transaction (RMB 7.73 billion). The group raised its full-year 2025 sales target from 2.71 million to 3.0 million units in July 2025 (Geely 2026). **CHECK: Tabelle nennt 16.6]**

The EV transition introduces a strategic dependency absent from BYD's model: Geely's electrification relies on platforms developed through Volvo Cars (separately listed) and the Smart JV (a 50:50 arrangement with Mercedes-Benz), neither of which ZGH controls unilaterally, rather than on in-house battery and platform production. The NEV-majority milestone shows this dependency has not constrained competitive pace: a 51.5% NEV share positions Geely ahead of most global OEMs, if still behind BYD's near-total electrification.

Balance-sheet developments warrant attention but do not signal distress. The gearing ratio rose to 22.2% (from 8.8% at end-2024), driven primarily by the ~RMB 9.1 billion ZEEKR acquisition of Lynk & Co, a consolidation that simplifies the brand structure rather than operational deterioration. Net cash of RMB 38.9 billion and operating cash flow of RMB 15.0 billion confirm solvency headroom. The announced privatisation of ZEEKR Technology Group (July 2025) further simplifies the listed portfolio and reduces minority-shareholder drag. **[CHECK: 3.4 nennt Geely-Gearing 19.8%]**

The overall assessment is asymmetric but has shifted from neutral to cautiously positive. A high-performing core in Geely Auto and Volvo Cars sustains the group, and the NEV-majority milestone and 102% adjusted profit growth indicate the transition is on schedule. The surrounding loss-making subsidiaries, Polestar and Lotus Technology, create contingent liabilities that, if crystallised simultaneously, would strain ZGH's leveraged balance sheet. That risk persists, but H1 2025 results show the core generating enough cash to absorb it for now.

5.4 Comparative Synthesis: The EV Transition as Structural Audit

The EV transition functioned as a structural audit of each firm's prior choices. SAIC faces the most acute pressure: the capability dependency documented in Section 4.4 has hardened into deficit, and the capital-light JV model inverted shared upside into shared liability without the operational control required to restructure it (Section 5.1). BYD occupies the strongest position, carrying no ICE legacy and absorbing the price war through its integrated supply chain better than any partner-dependent rival. Geely is the most asymmetric: a cash-generative core (51.5% NEV share, adjusted operating profit up 102%) offsets peripheral contingent liabilities for now, but their simultaneous crystallisation would strain the leveraged model. SAIC's architecture was not built for the speed the transition demands; BYD's was; Geely's has proved more adaptive than expected, at rising balance-sheet cost.

6. Generalizable Lessons

6.1 The Mode of Capability Acquisition Determines Durability Across Technology Cycles

SAIC, BYD, and Geely represent three modes of capability building: mandatory transfer through state-directed joint ventures, organic internal development, and cross-border acquisition. The JV regime generated quality spillovers but suppressed autonomous innovation through rent cannibalization (Bai et al., 2025; Howell, 2018), leaving SAIC without proprietary EV platforms. BYD's path-dependent accumulation of tacit knowledge produced non-imitable capabilities (Teece et al., 1997); Geely's voluntary, contractually governed acquisition avoided cannibalization and enabled co-development (Konda et al., 2022). The lesson: mandatory transfer buys short-run quality but displaces the autonomous innovation needed for the next cycle, while organic and voluntary modes preserve it, trading speed against depth against independence.

6.2 Governance Structure is a First-Order Determinant of Strategic Adaptability

In markets under rapid transition, the speed of capital reallocation is itself a competitive advantage, and it depends less on resources than on how control rights are distributed. SAIC's control is split across SASAC, a codified Party mandate, and 50/50 JV boards requiring bilateral consent, steering decisions toward objectives beyond firm value (Shleifer & Vishny, 1994; Fan et al., 2007) and delaying its EV response until 2025. BYD's consolidated founder control enabled the complete ICE exit in 2022 (Zhang et al., 2025); Geely trades acquisition agility against multi-brand coordination cost. The lesson: control concentrated with commercially oriented principals adapts faster than control distributed across political or joint-venture partners. When technological change outpaces a firm's decision cycle, governance becomes the binding constraint.

6.3 Capital Structure Determines Resilience When Markets Contract

The three financing models distribute risk differently under stress. SAIC's capital-light JV model captured dividends efficiently while markets grew, but when sales collapsed in 2024 it was bound to absorb losses it lacked the control to restructure (Section 3.1): the capital-light arrangement became a capital trap. BYD's self-financing model proved resilient, though sustained price competition erodes the internal cash it depends on. Geely's leveraged

acquisitions generated the highest return but created contingent liabilities through loss-making entities. Megginson and Netter (2001) supply the benchmark: mixed-ownership firms systematically underperform private peers. The lesson: risk-sharing captures upside in growth but exposes the firm in downturns where it lacks control; self-financing trades growth speed for resilience, leverage trades balance-sheet safety for strategic speed.

7. Conclusion

The three models examined in this paper, state-directed joint ventures (SAIC), organic vertical integration (BYD), and cross-border M&A (Geely), represent structurally distinct answers to the same challenge: how to build competitive capabilities at scale in a technology-intensive, rapidly evolving market. Across four analytical dimensions, the evidence converges on a single finding: governance structure is a first-order determinant of long-term value creation, not a second-order constraint to be adjusted after the strategy is set.

The cases bear this out in sequence. SAIC's JV model transferred production quality but suppressed the autonomous innovation EV competition requires, and state ownership left it unable to reallocate capital at the speed of the transition (Bai et al., 2025; Howell, 2018; Shleifer & Vishny, 1994). BYD's founder-led vertical integration, built ahead of state policy rather than in response to it, proved the most resilient (Whitfield & Wuttke, 2026), while Geely's acquisition model compressed the capability timeline at the cost of balance-sheet exposure contingent on Volvo's continued profitability (Balcet et al., 2012).

The broader implication extends beyond the automotive sector. In markets where technology cycles compress decision horizons, governance is not a variable firms optimize after their strategy is chosen; it is the mechanism through which any strategy is executed. Control rights concentrated with commercially oriented principals adapt faster to discontinuities, and capital structures that internalize rather than share risk remain resilient when partner assets decline. The Chinese automotive transition from 2020 to 2025 constitutes, in this sense, a large-scale natural experiment in the performance consequences of governance design under rapid technological change.

To-Dos:

☒ ~~Gedankenstriche minimieren (sieht zu KI aus)~~

☐ Lessons gegenprüfen

- ☐ Conclusion
- ☐ Checken, ob Einleitung zum Text passt
- ☐ Texte anpassen zu einem einheitlichen flüssigen Paper
- ☒ ~~Kürzen!!!!~~
- ☐ Literaturverzeichnis
- ☐ Autorennzuordnung
- ☐ Einheitliche Währung
- ☐ KI-Annex
- ☐ Ehrenwörtliche Erklärung
- ☐ Präsentation Slides checken & anpassen
- ☐ Alles gegenprüfen und Korrekturlesen
- ☒ Zitationen checken (vor allem Annual Reports):

(SAIC Motor, 2026), (BYD, 2026), (Geely, 2026) für Annual Report 2025 In Text
Referenzen (Author, Erscheinungsjahr).

(SAIC Motor, 2025), (BYD, 2025), (Geely, 2025) für Annual Report 2024

Nächstes Treffen Freitag 26.06. PEG Bib Raum 2 13.30 Uhr

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Note: Three entries require verification before submission: (1) Whitfield & Wuttke (2026) — confirm volume/issue against title page; (2) Yuan & Brasó Broggi (2025) — DOI 10.1080/00076791.2023.2247366 appears to be online-first, confirm final pagination; (3) Geely (2025b) — add HKEX document number if available.

- Whitfield & Wuttke (2026): Volume/Issue aus dem PDF bestätigen (ich habe „Vol. 4, Article 100054“ aus dem Wiki-Eintrag — bitte gegen das Titelblatt prüfen).
- Yuan & Brasó Broggi (2025): DOI 10.1080/00076791.2023.2247366 — das ist ungewöhnlich (2023-DOI für 2025-Paper), bitte kontrollieren ob das die Online-First-Version ist.
- Geely (2025b): Das Interim-Results-Announcement — falls ihr die genaue HKEX-Dokumentnummer habt, ergänzen; sonst reicht die URL.

KI Annex

Platform and Agent Architecture

This paper was researched and written with Claude Code (Anthropic), drawing on two models depending on task complexity: claude-sonnet-4-6 served as the primary model for the majority of research and drafting tasks, while claude-opus-4 was used selectively for more demanding analytical work, including cross-case comparisons, developing the paper's central argument, and evaluating the logical consistency of multi-source claims. Rather than a single general-purpose assistant, the setup relied on a layered system of specialized agents: wiki-ingest for structured ingestion of PDFs and web sources, wiki-query for extracting specific information from the accumulated knowledge base, wiki-lint for health-checking the wiki for broken links, unresolved contradictions, and stale data, verify-claims for tracing individual factual claims back to primary sources before incorporating them into drafts, and find-references for identifying additional supporting literature.

Shared Repository and Collaboration

The three group members worked within a shared git repository, with each case in a dedicated wiki subfolder (research/wiki/saic-motor/, research/wiki/byd/, research/wiki/geely/) and a shared layer at research/wiki/shared/ for cross-case comparison data, the EU Implementing Regulation, and the IEA Global EV Outlook. The rule was simple: git pull before every session, git push after. Claude Code had read access to all three wikis simultaneously, which enabled direct cross-case comparisons without requiring group members to manually share data. One merge conflict arose during the project (two members pushing to the same file simultaneously) and was resolved using git stash to preserve local changes before pulling the remote state.

Source Quality System and the Writing guide

Early in the project, a formal source-quality rating was created for each case wiki. Every source was assessed on two dimensions: journal quality (Q1 = ABS A* such as AER, QJE, JFE; Q2 = ABS A; Q3 = ABS B; down to Q5 = unreviewed web material) and direct relevance to the respective case (R1 = company explicitly named as a case study; R2 = Chinese automotive industry; R3 = theoretical framework; R4 = background context only). This rating was consulted at the start of every writing session to ensure that empirical claims were anchored in the highest-quality available source and that lower-ranked sources were not used in positions they could not support.

To keep the drafting phase efficient across multiple sessions, a writing guide was compiled for each case: a structured reference document consolidating all verified data points, the correct source assignment for each argument, and the applicable citation conventions. Rather than re-querying the wiki each time a figure or reference was needed, the writing guide provided a single verified starting point and ensured that citation formatting remained consistent across sessions and across group members drafting in parallel.

Commands and Prompts That Were Particularly Helpful

Beyond the agent invocations, several prompting patterns proved consistently effective. Asking Claude Code to act as a "critical academic reviewer", a critical supervisor who identifies errors and inconsistencies rather than validates the existing text, consistently produced sharper feedback than generic review requests. Asking explicitly to verify a single claim against its source before incorporating it (rather than trusting the wiki summary) was the approach that caught the three significant data errors described below. For structuring arguments, providing Claude Code with the source-rating file and asking it to "build the argument using only Q1 and Q2 sources as primary anchors" prevented lower-quality sources from appearing in positions they could not support.

How Helpful It Was, and Where It Had Limits

The platform was most valuable for handling research volume that would have been unmanageable manually: across all three cases, more than 30 academic papers and multiple Annual Reports totalling well over 1,500 pages were ingested, indexed, and made queryable within a persistent knowledge base. Three genuine data errors were caught before entering the paper: a provisional EU tariff figure (36.3%) corrected to the final regulatory figure (35.3%), a net profit value inflated by a factor of ten through a currency-unit ambiguity in a secondary source, and an EPS figure based on a wrong share count, all corrected by tracing back to the PwC- or Ernst & Young-audited annual reports. The platform's persistence was also practically important: one session was lost to a hardware shutdown, and the entire research base was recoverable from the wiki and a cross-session memory system without rebuilding from scratch.

Limitations were real. The platform cannot access paywalled journals and had no capability to process Geely's Annual Report PDFs, which were not machine-readable in the working environment, so the Geely case relied more heavily on web research inputs as a result. When using more capable models or working through research-intensive sessions, token limits were reached relatively quickly, requiring waiting periods before work could continue. AI-generated formulations required active critical review; early drafts contained overclaims and

framing errors that the authors identified and corrected. Analytical judgment, argumentation, and all interpretive decisions remained entirely with the authors.

Setup and Costs

Setup required installing the Claude Code CLI, git, and the apm package manager, then running apm install once to deploy all agents, skills, and MCP server configurations. While the technical installation itself was completed within one session, familiarizing ourselves with the platform. Understanding how the agent system, wiki workflow, and MCP servers interact, took several additional hours. All team members used the Claude Pro subscription at approximately €21 per month; no additional per-query charges apply. No paid third-party data services were used.

Declaration of Authorship

All arguments, analytical claims, structural decisions, and conclusions in this paper are the authors' own. Claude Code was used as a research infrastructure tool, a source-quality enforcer, and a critical editorial reviewer, not as a ghostwriter. No passage was incorporated without author review and, where necessary, revision.

Phase	AI support	Own contribution
Literature research	Parallel web searches across multiple angles simultaneously; identification of relevant academic sources; source quality rating by journal rank (ABS A* down to unranked)	Literature search strategy within assigned cases and research questing; selection of specific academic sources, obtaining all papers

Statutory Declaration

“We herewith declare that we have composed the present term paper ourselves and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The term paper in the same or similar form has not been submitted to any examination body and has not been published. This thesis was not yet, even in part, used in another examination or as a course performance.”

Place, Date: Frankfurt am Main, 29.06.2026

A handwritten signature in black ink, appearing to be 'Jant', written over a horizontal line.

Signature: